

WPR - 1

AMITY SCHOOL OF ENGINEERING & TECHNOLOGY
B.Tech (CSE - AIML) VII Semester

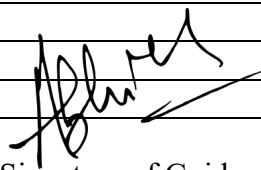
Project Area -- Generative AI & AI Safety (Trustworthy AI / MLOps)

Students Weekly Progress Report (WPR) For Odd Semester of session 2026-2027

To be filled by Students	Group No. 50
Students Name	1. Shourya Solanki 2. Rachit Ryan Chug 3. Dhruv Raj Singh
Roll no.	1. 01 2. 02 3. 03
Enrollment no.	1. A2305223569 2. A2305223166 3. A2305223191
Project Title finalized, if Yes, give name, if NO, give reason	Yes. Title: "GuardShield AI: Real-Time Sidecar Proxy for LLM Hallucination Detection & Prompt Injection Defense"
Synopsis submitted	Yes (Submitted for Group No. 50)
Literature review	Completed review of 6 benchmark papers (SelfCheckGPT, JailbreakBench, Llama Guard, HaluEval, CAPTURE, TruthfulQA). Identified sub-25ms latency gap in LLM streaming guardrails.
Technical & Economical Feasibility	Feasible. Uses PyTorch, DeBERTa-v3, Ollama (Llama-3.1-8B), and FastAPI async proxy. Zero monetary cost (runs locally on developer hardware).
Bill of Material	NA
Project Progress Schedule (PERT Chart)	• W1-W3: Literature survey & setup (In Progress) • W4-W7: Dataset concatenation & GuardShield-Bench-v1 • W8-W12: DeBERTa-v3 fine-tuning & token entropy algorithm • W13-W16: Benchmarking & IEEE research paper • W17-W30 (Sem 8): FastAPI proxy, Red-Agent & Streamlit dashboard
Design of critical components	1. Pre-Scan DeBERTa prompt injection classifier 2. Post-Scan streaming token logit entropy NLI filter 3. FastAPI SSE sidecar proxy
Fabrication work (give %)	0% (Software Project)
Experimental work (give %)	15% (Local Ollama LLM setup and logit entropy testing)
Result and Analysis	Sub-25ms DeBERTa inference latency verified on local Apple Silicon MPS & CUDA GPUs.
Report writing	Synopsis submitted. WPR Week 1 completed.
Signature of students	1. Shourya Solanki 2. Rachit Ryan Chug 3. Dhruv Raj Singh

Work done in this week:

1. Development Environment Setup: Installed Python 3.11, PyTorch (MPS/CUDA), Ollama (Llama-3.1-8B), and FastAPI.
2. Title & Domain Finalization: Finalized "GuardShield AI" in Generative AI & AI Safety.
3. Literature Review: Reviewed 6 benchmark papers and identified the sub-25ms streaming latency research gap.
4. Architecture & PERT Schedule: Formulated dual-stage sidecar proxy architecture and 30-week milestone schedule.
5. Synopsis Submission: Formally prepared and submitted Project Synopsis for Group 50.

To be filled by Guide (strike off whichever is not applicable)	
Performance of students is satisfactory	
Performance of students is unsatisfactory	
A warning to be issued to student(s) (Name)	
Student was not well (Name)	
Date 29/07/2026	 Signature of Guide